



CSE 4208 : Computer Graphic
Laboratory

3D Hover Vehicle Simulator Game

Presented To:

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Outline

- Introduction
- Objectives
- Tools Used
- Textures
- Snapshots
- Video Demonstration
- Conclusion

Introduction

- A 3D game environment implemented using Modern OpenGL 3.3.

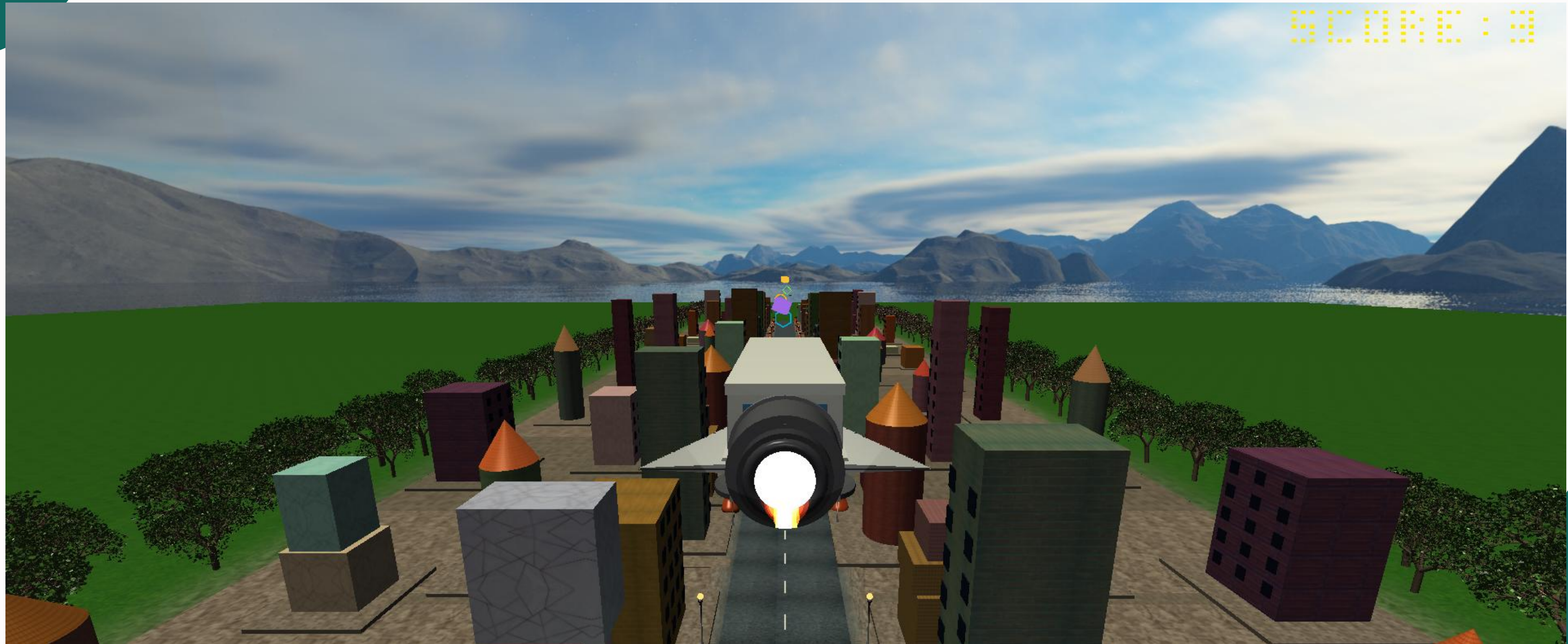


Figure 1: View of the complete Game environment

Objectives

- Build a visually stunning 3D Hover vehicle game that incorporate realistic structures , physics laws.
- To implement multiple lighting models, including directional, point, spotlights , to replicate real-world lighting conditions.
- To use high-quality textures and ensure their seamless integration with the 3D models..
- To integrate interactive camera controls, enabling users to explore the 3D scene dynamically

Tools Used

- OpenGL(3.3): Rendering 2D and 3D graphics.
- Glad: Glad is a straightforward and simple loader generator that manages OpenGL function pointers, enabling easy and efficient management of OpenGL
Not optimized for real-time processing
- GLFW (Graphics Library Framework): Window creation and handling input (keyboard, mouse).

Textures



Figure 2 : Stone wall

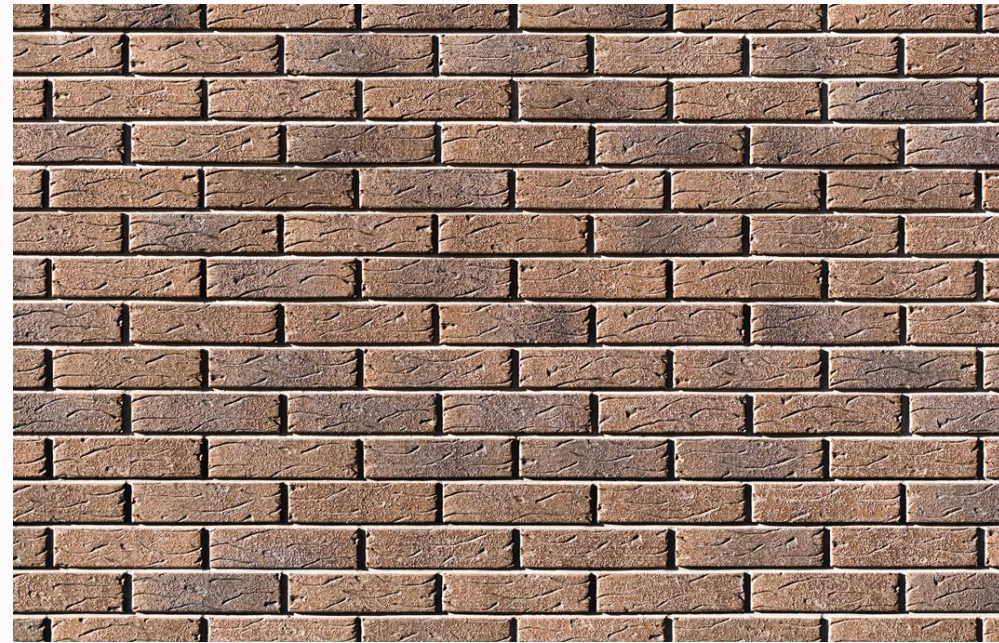


Figure 3 : brick wall

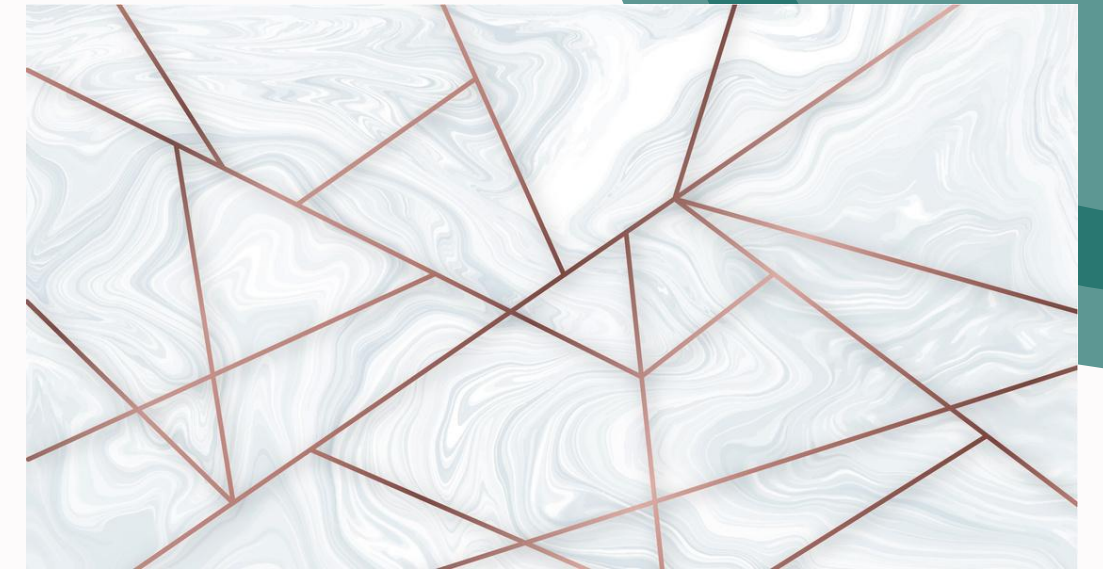


Figure 4 : Marble Wall



Figure 5: tile



Figure 6: earth tone tile



Figure 7: Carpet tile

Textures(Contd.)



Figure 8 : Road



Figure 9 : Tree bark



Figure 10 : Tree leaf

Textures(Contd.) - skybox

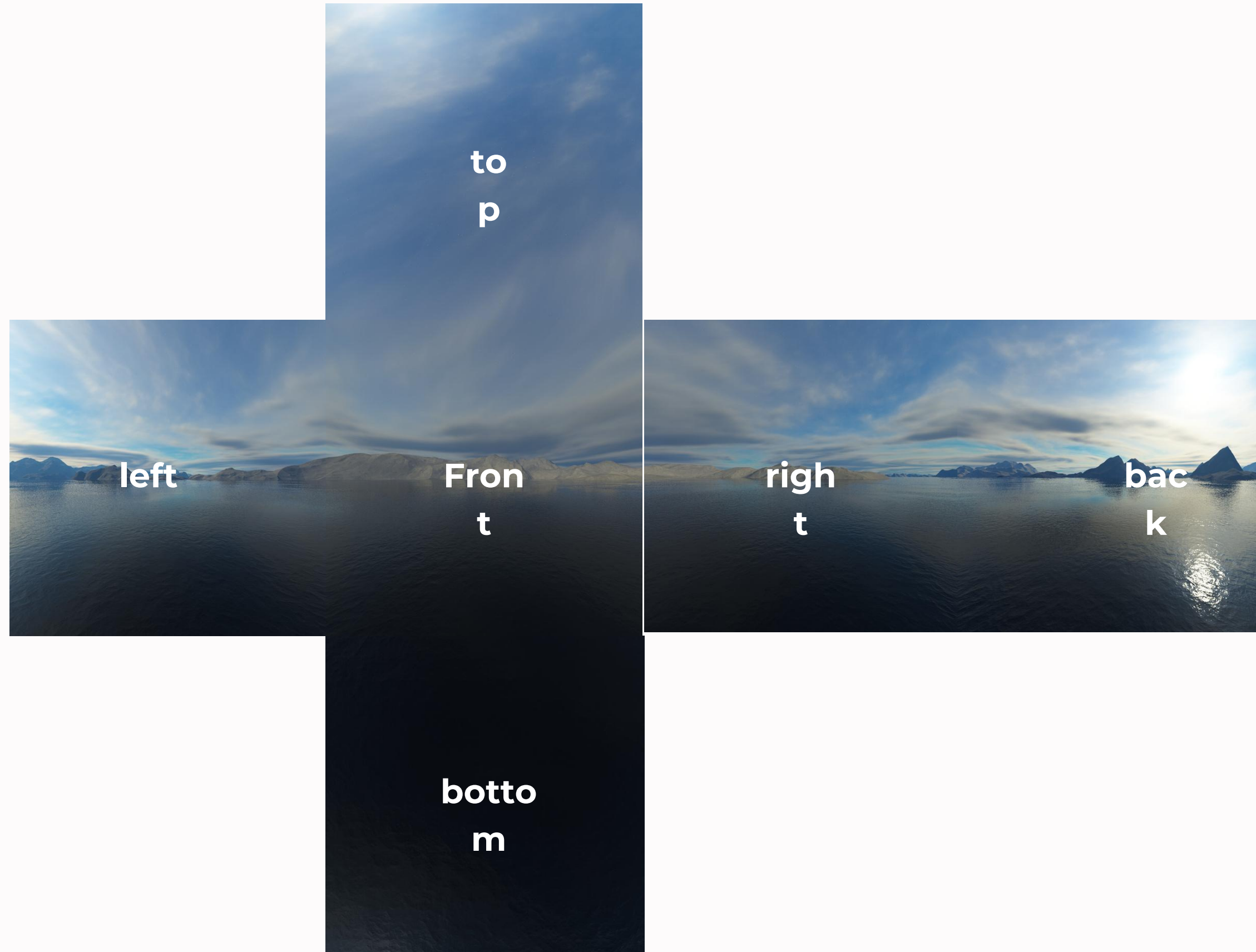
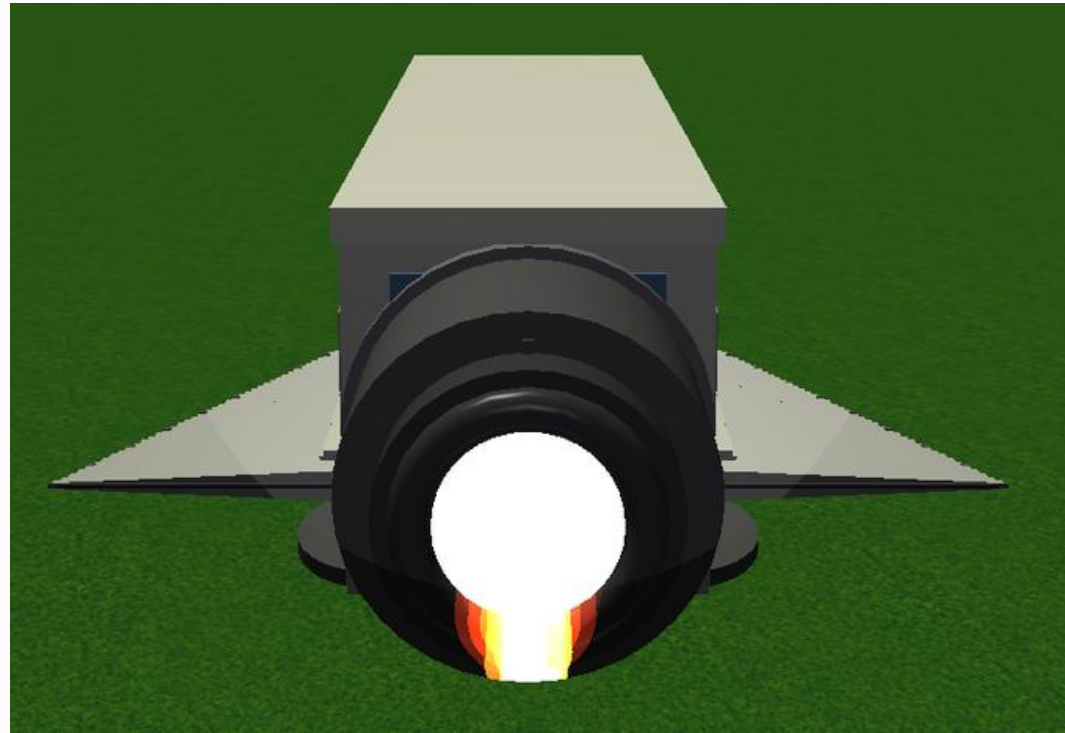


Figure 10 : 6 skybox images(2048x2048)

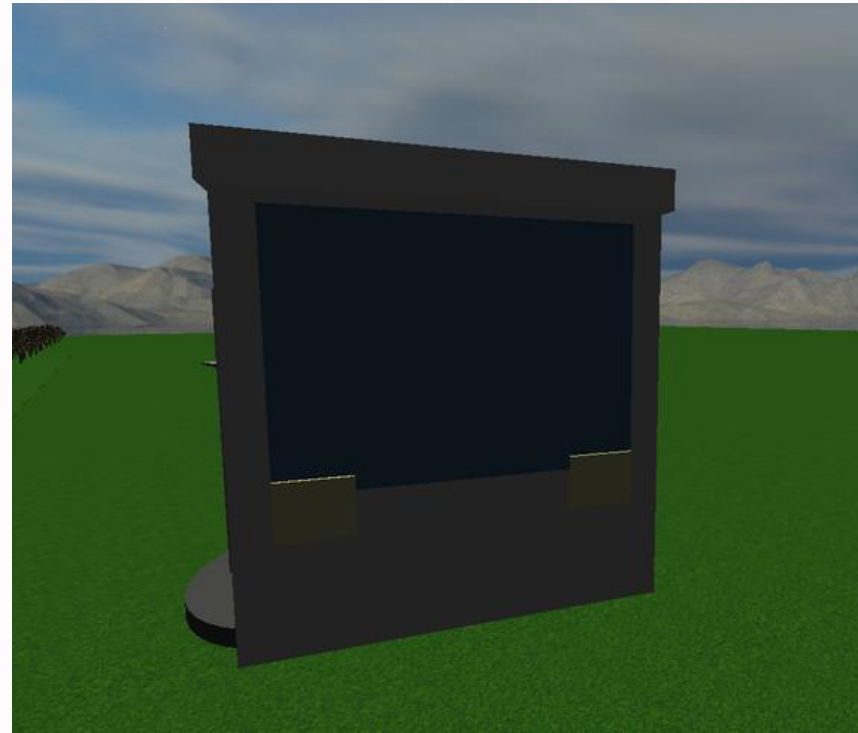


Figure 11 : field grass

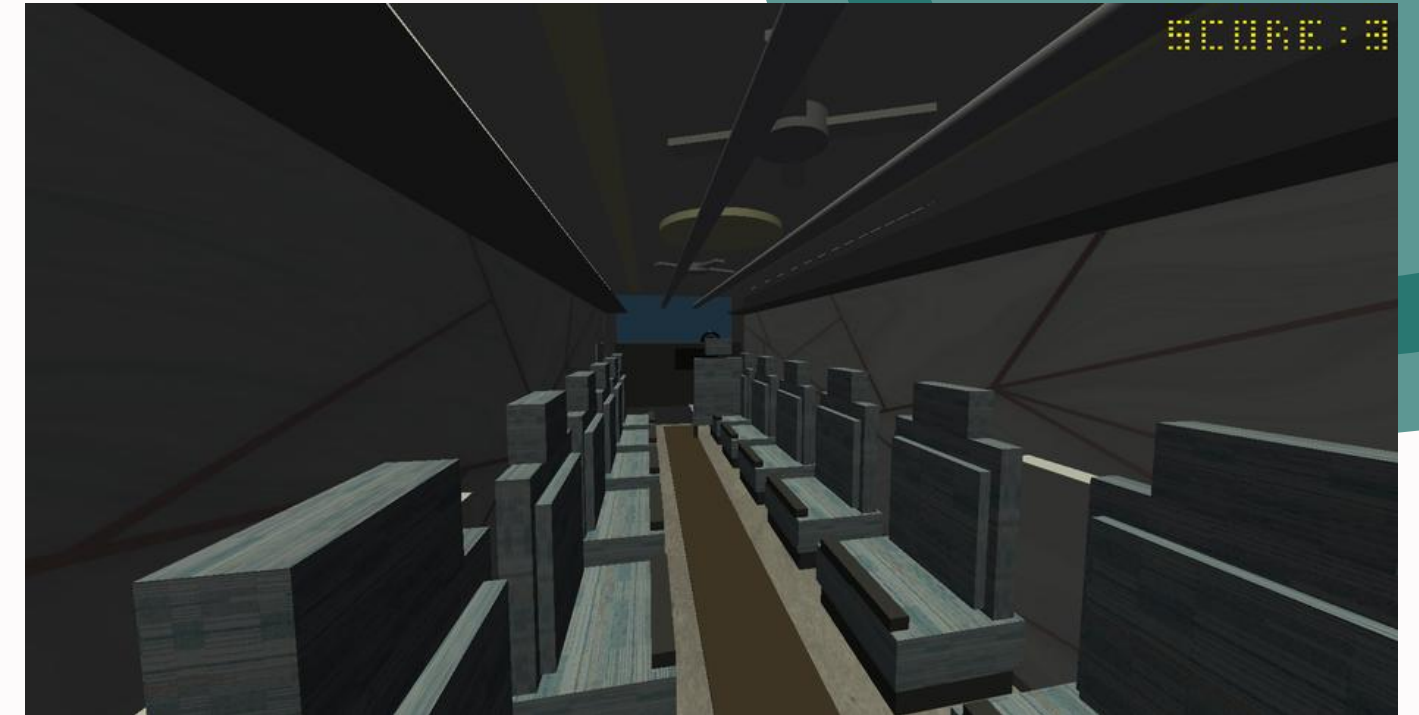
3D Objects - Bus



Back



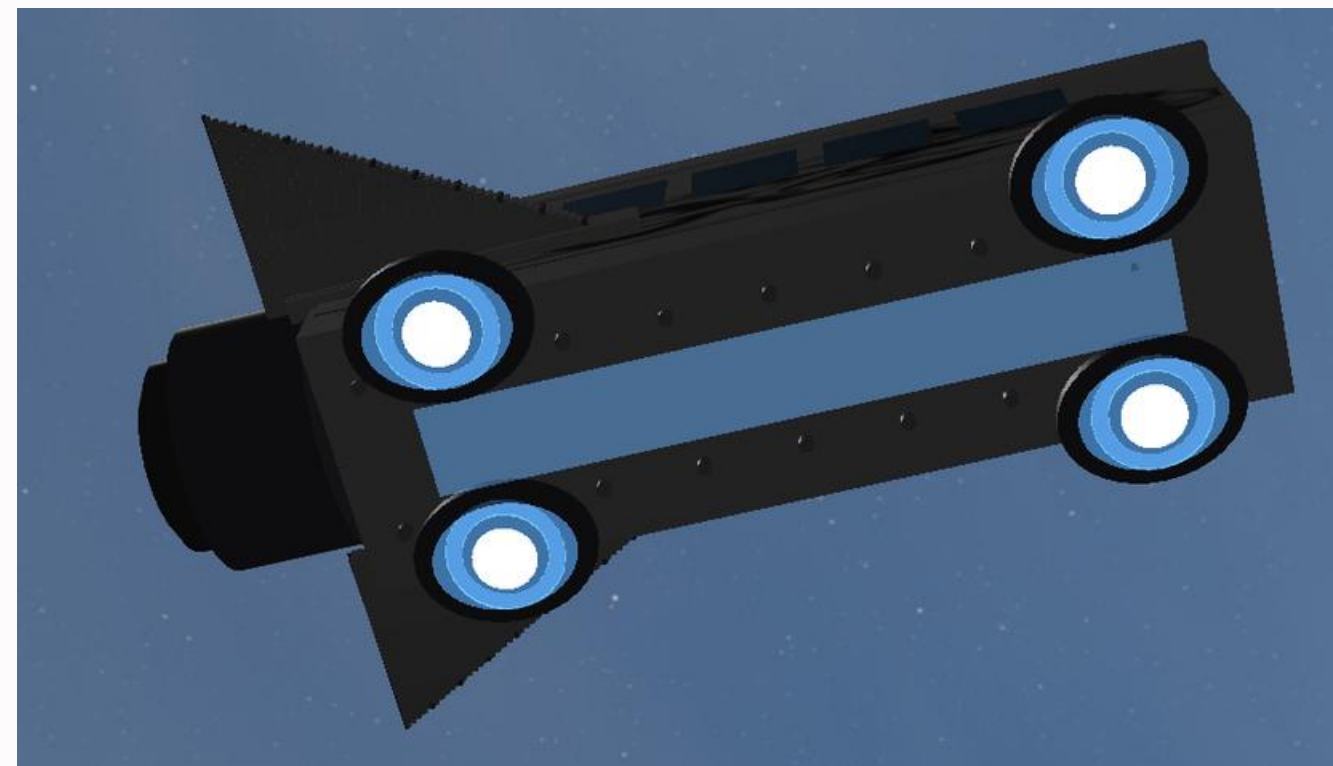
Front



Inside of the bus



Side



Bottom

- Primitives used:**
- Cube
 - Torus
 - Cylinder

3D Objects - Buildings & Roads



Buildings



Warehouse



pitch Road

3D Objects - trees



Tree with leaf texture mapping

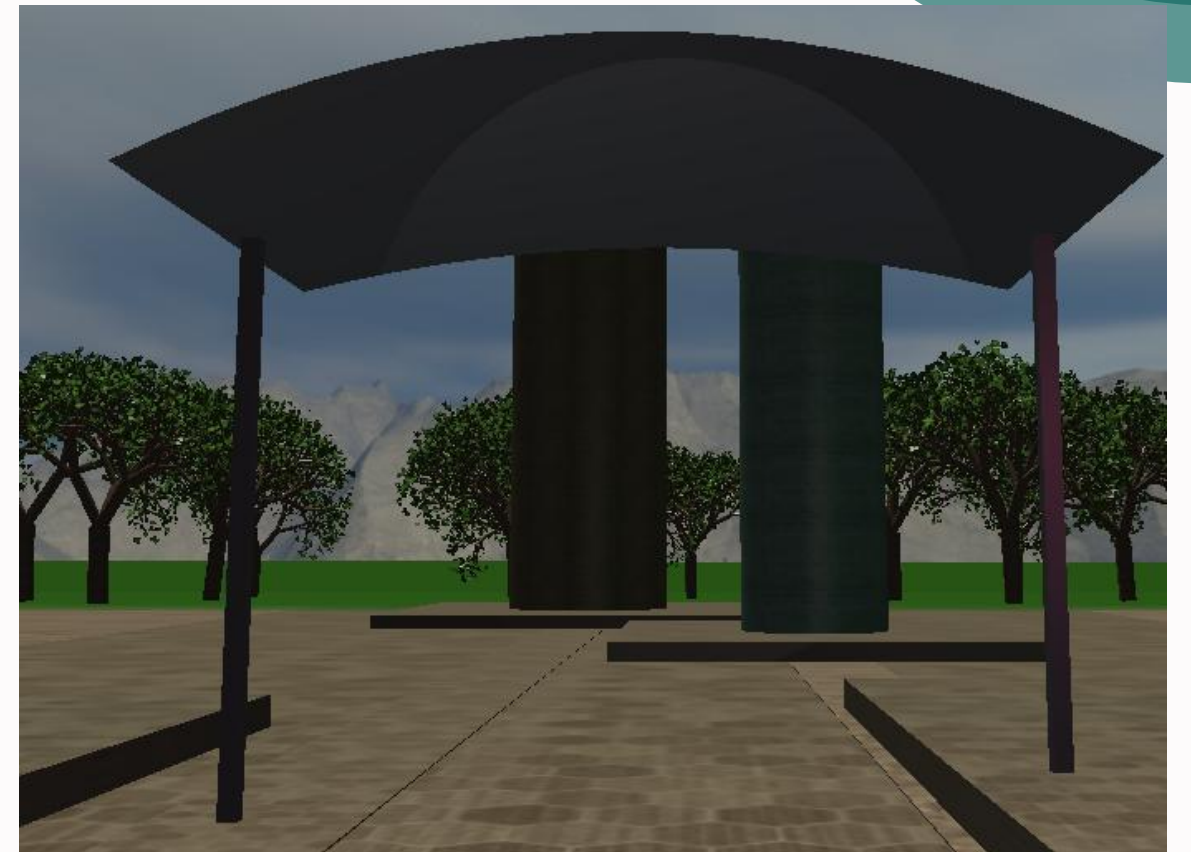
3D Objects - Curves



Vase(Bezier curve)



**street lamp (
Spline Surface)**

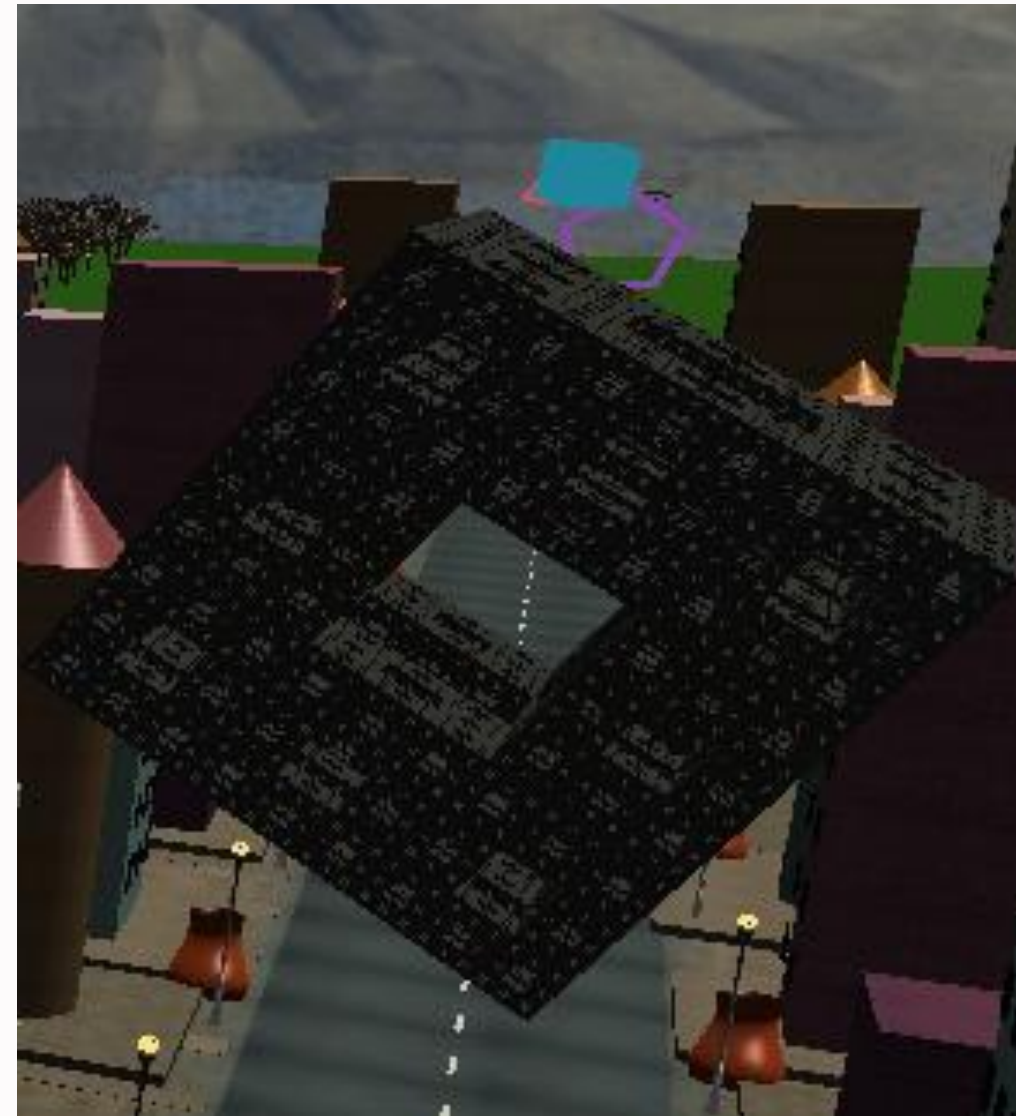


Bus Stop(ruled Surface)

3D Objects - Fractals



**Forest lining(recursive branching
fractals)**



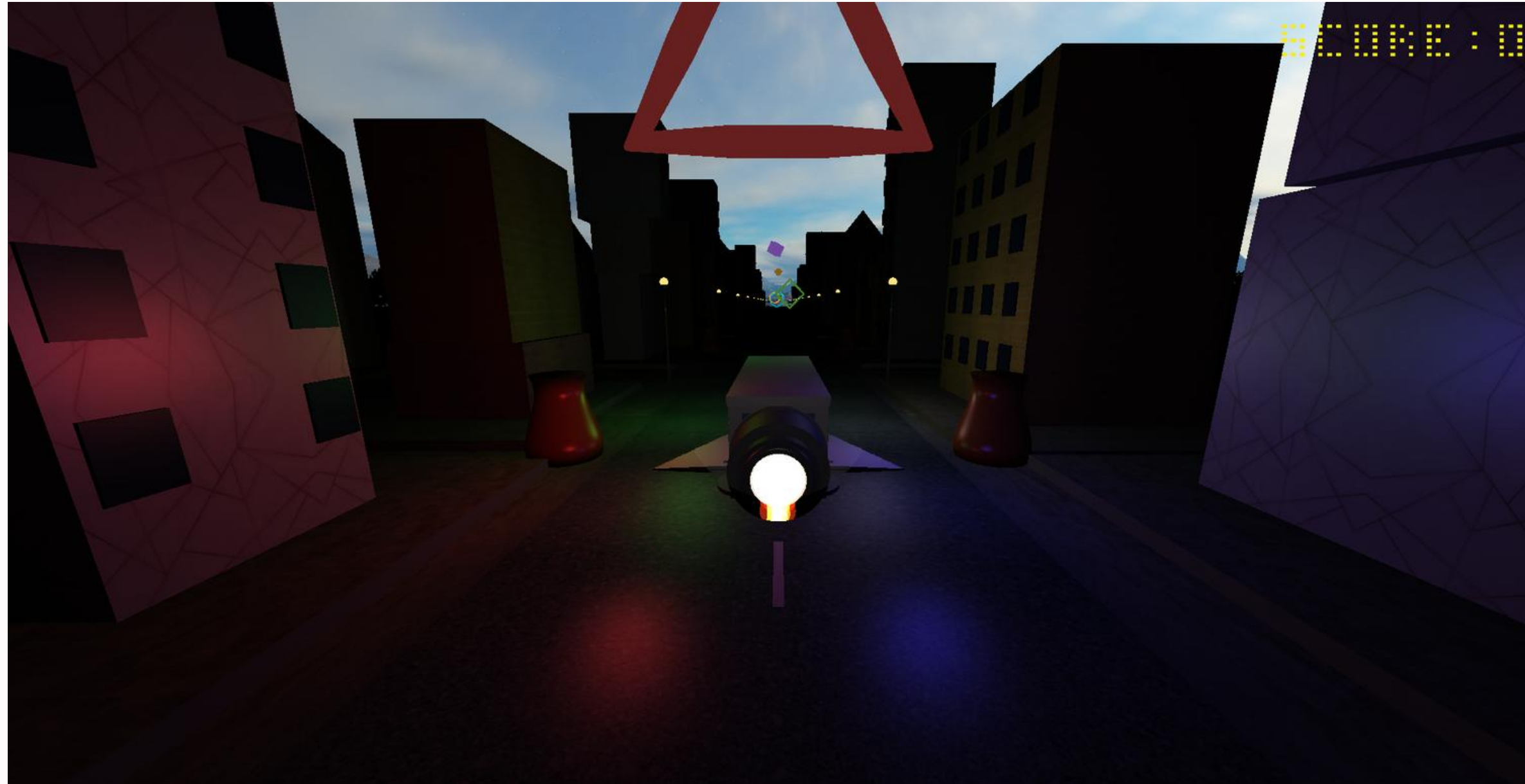
**Menger sponge(Iterative
Subdivision Fractal)**

Lighting & Shading



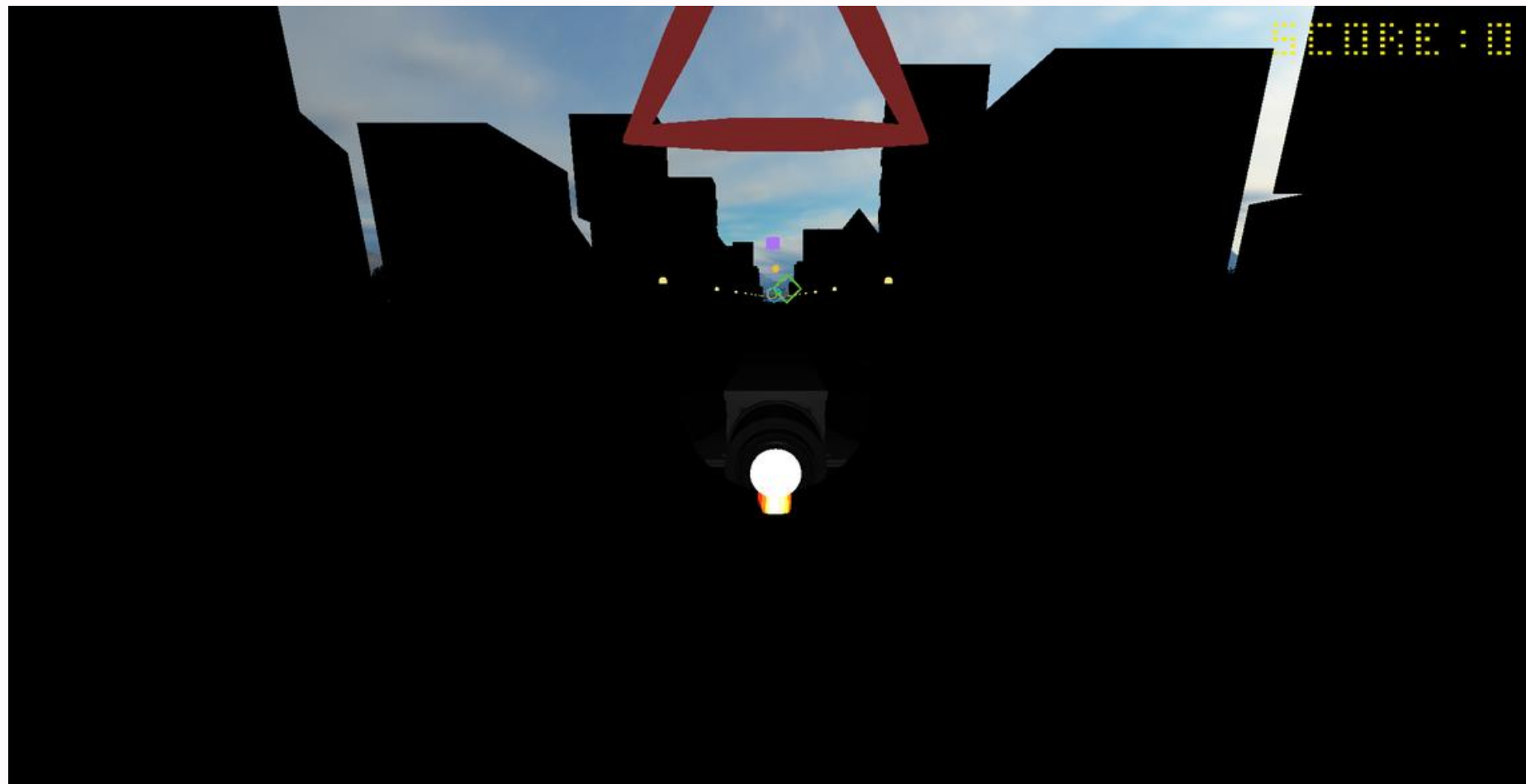
**Directional Light , Point lights, spotlight
emissive glow , ambient light ON**

Lighting & Shading



Directional Light OFF

Lighting & Shading

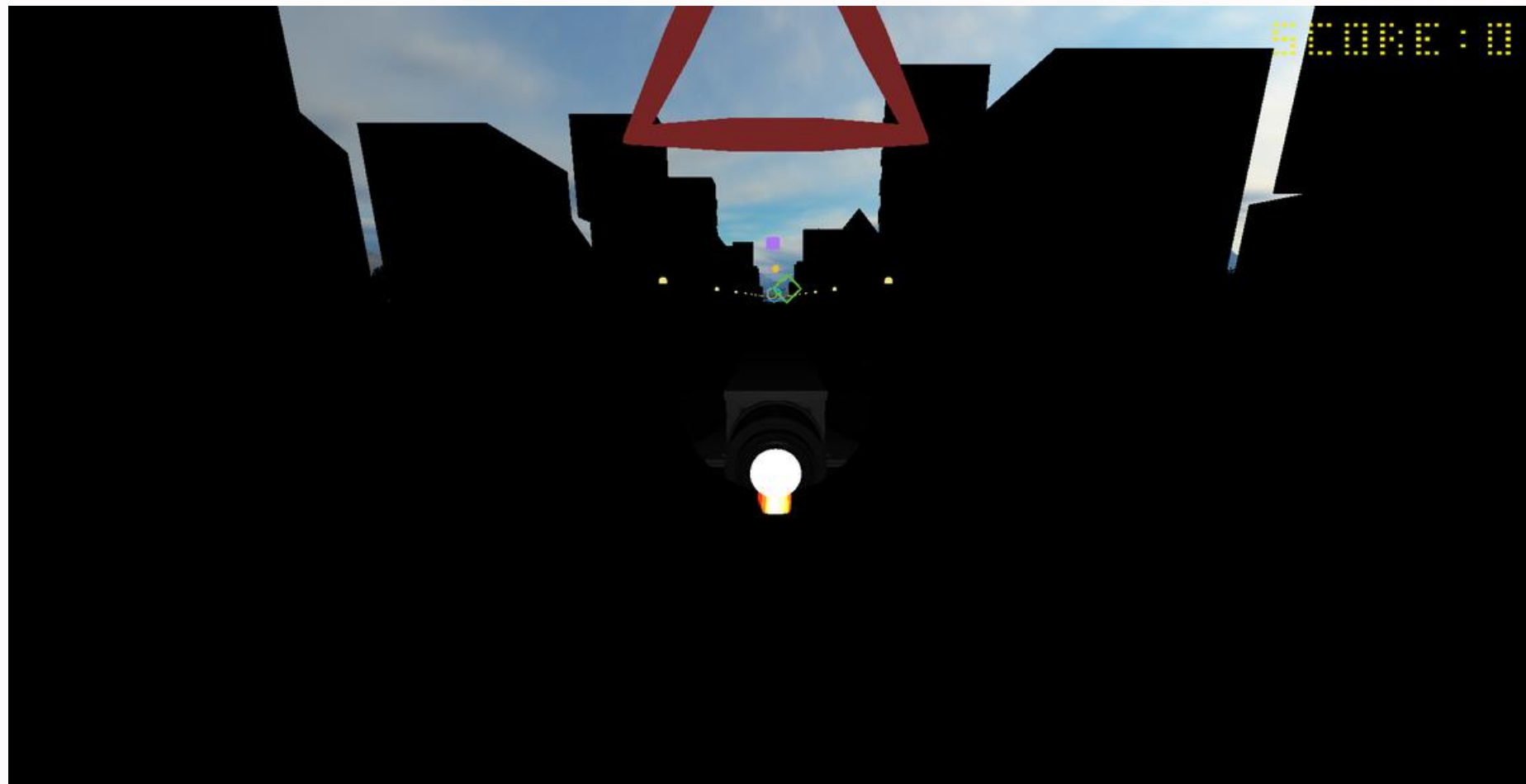


Directional Light OFF , Point Light Off



Emissive Only

Lighting & Shading



Directional Light OFF , Point Light Off



Emissive Only

Video Demonstration



Conclusion

- Successful Development of 3D City Model
- Application of Advanced OpenGL Concepts
- Learning and Integration of External Tools.

The image features a large, dark teal circle in the center. Overlaid on this circle is the text 'THANK YOU' in a bold, yellow, sans-serif font. Surrounding the central circle are several other geometric elements: a small dark teal circle at the top left, a medium dark teal circle at the top right, a light green circle with a white center at the top right, a dark teal circle with a white border at the bottom left, and a light green arc at the bottom left. A dark teal arc is also visible at the bottom right.

THANK YOU